5. If $2\sin 2\theta = \sqrt{3}$, then the value of θ is

(1) 90° (2) 30° (3) 45° (4) 60°

6. The value of $3\sin 70°\sec 20° + 2\sin 49°\sec 51°$ is

(1) 2 (2) 3 (3) 5 (4) 6

7. The value of $\dfrac{1-\tan^2 45°}{1+\tan^2 45°}$ is

(1) 2 (2) 1 (3) 0 (4) $\dfrac{1}{2}$

8. The value of $\operatorname{cosec}(70°+\theta) - \sec(20°-\theta) + \tan(65°+\theta) - \cot(25°-\theta)$ is

(1) 0 (2) 1 (3) 2 (4) 3

9. The value of $\tan 1° \tan 2° \tan 3° \ldots \tan 89°$ is

(1) 0 (2) 1 (3) 2 (4) $\dfrac{\sqrt{3}}{2}$

10. Given that $\sin\alpha = \dfrac{1}{2}$ and $\cos\beta = \dfrac{1}{2}$, then the value of $\alpha+\beta$ is

(1) 0° (2) 90° (3) 30° (4) 60°

Points to Remember

- Trigonometric ratios are

$\sin\theta = \dfrac{\text{opposite side}}{\text{hypotenuse}}$	$\operatorname{cosec}\theta = \dfrac{\text{hypotenuse}}{\text{opposite side}}$
$\cos\theta = \dfrac{\text{adjacent side}}{\text{hypotenuse}}$	$\sec\theta = \dfrac{\text{hypotenuse}}{\text{adjacent side}}$
$\tan\theta = \dfrac{\text{opposite side}}{\text{adjacent side}}$	$\cot\theta = \dfrac{\text{adjacent side}}{\text{opposite side}}$

- Reciprocal trigonometric ratios

$\sin\theta = \dfrac{1}{\operatorname{cosec}\theta}$ $\quad \cos\theta = \dfrac{1}{\sec\theta}$ $\quad \tan\theta = \dfrac{1}{\cot\theta}$

$\operatorname{cosec}\theta = \dfrac{1}{\sin\theta}$ $\quad \sec\theta = \dfrac{1}{\cos\theta}$ $\quad \cot\theta = \dfrac{1}{\tan\theta}$

- Complementary angles

$\sin\theta = \cos(90°-\theta)$ $\quad \operatorname{cosec}\theta = \sec(90°-\theta)$

$\cos\theta = \sin(90°-\theta)$ $\quad \sec\theta = \operatorname{cosec}(90°-\theta)$

$\tan\theta = \cot(90°-\theta)$ $\quad \cot\theta = \tan(90°-\theta)$

6 Trigonometry.indd 240 02/12/2022 21:48:22